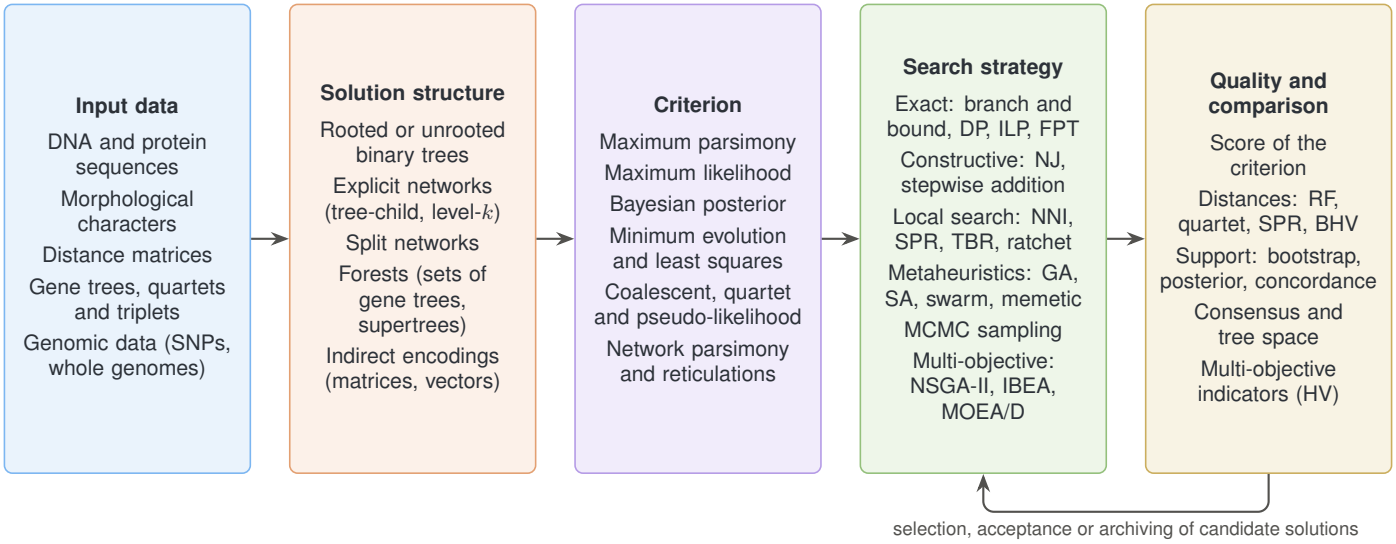
Input data
DNA and protein sequences
Morphological characters
Distance matrices
Gene trees, quartets and triplets
Genomic data (SNPs, whole genomes)
Solution structure
Rooted or unrooted binary trees
Explicit networks (tree-child, level-k)
Split networks
Forests (sets of gene trees, supertrees)
Indirect encodings (matrices, vectors)
Criterion
Maximum parsimony
Maximum likelihood
Bayesian posterior
Minimum evolution and least squares
Coalescent, quartet and pseudo-likelihood
Network parsimony and reticulations
Search strategy
Exact: branch and bound, DP, ILP, FPT
Constructive: NJ, stepwise addition
Local search: NNI, SPR, TBR, ratchet
Metaheuristics: GA, SA, swarm, memetic
MCMC sampling
Multi-objective: NSGA-II, IBEA, MOEA/D
Quality and comparison
Score of the criterion
Distances: RF, quartet, SPR, BHV
Support: bootstrap, posterior, concordance
Consensus and tree space
Multi-objective indicators (HV)
selection, acceptance or archiving of candidate solutions